Q1 - 24 June - Shift 1

Identify the pair of physical quantities which have different dimensions :

(A) Wave number and Rydberg's constant

(B) Stress and Coefficient of elasticity

(C) Coercivity and Magnetisation

(D) Specific heat capacity and Latent heat

Space for your notes:

Q2 - 24 June - Shift 2

Identify the pair of physical quantities that have same dimensions :

(A) velocity gradient and decay constant

(B) wien's constant and Stefan constant

(C) angular frequency and angular momentum

(D) wave number and Avogadro number

Space for your notes:

Q3 - 26 June - Shift 1

An expression for a dimensionless quantity P is given by $P = \frac{\alpha}{\beta}\log_e\left(\frac{kt}{\beta x}\right)$; where α and β are constants, x is distance ; k is Boltzmann constant and t is the temperature. Then the dimensions of α will be :

(A) $[M^0L^{-1}T^0]$ (B) $[ML^0T^{-2}]$

(C) $[MLT^{-2}]$ (D) $[ML^2T^{-2}]$

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Q4 - 26 June - Shift 2

The dimension of mutual inductance is :

(A) $[ML^2T^{-2}A^{-1}]$ (B) $[ML^2T^{-3}A^{-1}]$

(C) $[ML^2T^{-2}A^{-2}]$ (D) $[ML^2T^{-3}A^{-2}]$

Space for your notes:

Q5 - 27 June - Shift 2

The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be

(A) $[ML^{-1}T^{-1}]$ (B) $[ML^{-1}T^{-2}]$

(C) $[ML^2T^{-1}]$ (D) $[M^{-1}L^3T^0]$

Space for your notes:

Q6 - 27 June - Shift 2

If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time ?

(A) RC (B) $\frac{L}{R}$

(C) $\sqrt{LC}$ (D) $\frac{L}{C}$

Space for your notes:

Q7 - 29 June - Shift 1

In Vander Waals equation $\left[P+\frac{a}{V^2}\right][V-b]=RT$; P is pressure, V is volume, R is universal gas constant and T is temperature. The ratio of constants $\frac{a}{b}$ is dimensionally equal to :

(A) $\frac{P}{V}$ (B) $\frac{V}{P}$

(C) PV (D) PV^3

Space for your notes:

Answer Key

Q1 (D)	**Q2** (A)	**Q3** (C)	**Q4** (C)
Q5 (A)	**Q6** (D)	**Q7** (C)	

Q1 (D)

$$S = \frac{Q}{m\Delta T} = \frac{J}{Kg^{\circ}C}$$

$$L = \frac{Q}{m} = \frac{J}{Kg}$$

Q2 (A)

$$\text{Velocity gradient } = \frac{dV}{dx} = \frac{1}{S}$$

$$\lambda = \frac{1}{S}$$

Q3 (C)

$$P = \frac{\alpha}{\beta}\log_e\left(\frac{kt}{\beta x}\right)$$

$$\frac{kt}{\beta x} = 1 \Rightarrow \beta = \frac{kt}{x} = \frac{ML^2T^{-2}}{L}$$

$$\left(\because E = \frac{1}{2}kt\right)$$

As P is dimensionless

$$\Rightarrow [\alpha] = [\beta] = [MLT^{-2}]$$

Q4 (C)

e_2 : induced emf in secondary coil

i_1 : Current in primary coil

M : Mutual inductance

$$e_2 = -M\frac{di_1}{dt}$$

$$M = -\frac{e_2}{\frac{di_1}{dt}}$$

$$[M] = \frac{[e_2]}{\left[\frac{di_1}{dt}\right]} = \frac{\left[\frac{W}{q}\right]}{\left[\frac{di_1}{dt}\right]} = \frac{\left[ML^2T^{-2}\right]}{\frac{[AT]}{\left[AT^{-1}\right]}}$$

$$= [ML^2T^{-2}A^{-2}]$$

Q5 (A)

Pascal second

$$\frac{F}{A}t = \frac{MLT^{-2}}{L^2}T = ML^{-1}T^{-1}$$

Q6 (D)

$\left(\frac{L}{C}\right)$ does not have dimension of time.

RC, $\frac{L}{R}$ are time constant while $\sqrt{LC}$ is reciprocal of angular frequency or having dimension of time.

Q7 (C)

By principle of homogenity

$$[P]=\left[\frac{a}{v^2}\right] \text{ and } [b]=[v]$$

$$\Rightarrow\left[\frac{a}{b}\right]=[PV] \text{ (C)}$$